

Material Safety Data Sheet

Company's Name and Address:

Hunan Haili Chemical Industry Co., Ltd.

No.399, Furong Road (M), Changsha, Hunan, China.

1. Product and company identification

1.1 Product trade name: Carbosulfan 20% EC

1.2 Product code: 3701

1.3 Emergency telephone number and fax (of the supplier):

2. Composition/Information on Ingredients

2.1 Common name or chemical name or generic name:

Carbosulfan 20% EC

2.2 Synonyms (if any): Marshal

2.3 CAS number: 55285-14-8

2.4 Ingredients contributing to the hazard: carbosulfan

2.5 Chemical nature: Carbamate insecticide

2.6 Components contributing to the hazard or impurities: contributing to the hazard:
Bis[(2,3-dihydro-2,2-dimethyl-7-benzofranyl) methylcarbamate] sulfide;
Bis(di-n-butylamino)trisulfide; Carbofuran; Bis(di-n-butylamino)tetrasulfide;
Bis(di-n-butylamino)pentasulfide; Organic solvent; Emulsifier; etc.

2.7 Concentration or concentration range: 20%

2.8 Classification and hazard labeling of components or impurities: II

3. Hazards identification

3.1 Adverse human health effects:

Effects from overexposure may result from either swallowing, inhaling or coming into contact with the skin or eyes. Conditions of increased temperature and humidity facilitate skin absorption of this product, and therefore, promotes increased toxicity.

3.2 Environmental effects:

Carbosulfan is rapidly degraded in neutral soil (half-life < 5 days) with the degradation rate increasing as pH decreases. A major breakdown product is carbofuran, which degrades more slowly (half-life approx. 50 days). Carbosulfan is hydrolytically unstable in acid, with stability increasing with increasing pH. The bioconcentration potential of carbosulfan is low with a Log Pow of 3.3, and a measured BCF in fish of 990. Carbosulfan and its major breakdown product are unlikely to leach into groundwater.

3.3 Physical and chemical hazards:

3.4 Specific hazards:

3.5 Main symptoms: Symptoms of overexposure include headache, light-headedness, weakness, abdominal cramps, nausea, excessive salivation, perspiration, blurred vision, tearing, pin-point pupils, blue skin color, convulsions, tremor and coma.

3.6 Classification of the product: EC

3.7 Emergency overview:

4 First-aid measures

4.1 Inhalation: Remove to fresh air. If breathing difficulty or discomfort occurs and persists, see a medical doctor. If breathing has stopped, give artificial respiration and see a medical doctor

immediately.

4.2 Skin contact: Remove contaminated clothing and thoroughly wash with soap and water. If irritation occurs and persists, contact a medical doctor.

4.3 Eye contact: Flush with water for at least 15 minutes. If irritation occurs and persists, contact a medical doctor.

4.4 Ingestion: Rinse mouth with water. Dilute by giving 1 or 2 glasses of water. Do not induce vomiting. Never give anything by mouth to an unconscious person. See a medical doctor immediately.

4.5 Important symptoms (detail of symptoms and effects should be given under heading 11):

4.6 Protection of first-aiders:

4.7 Notes to a physician: This product is moderately toxic if swallowed or inhaled, and slightly toxic to absorbed through the skin. It is moderately irritating to the skin and eyes. Carbosulfan is a reversible cholinesterase inhibitor. Atropine sulfate is antidotal. Support respiration as needed with removal of secretions, maintenance of a patent airway and, if necessary, artificial ventilation. If cyanosis is absent: Adults - start treatment by giving 2 mg atropine intravenously or intramuscularly, if necessary, and repeat with 0.4 - 2.0 mg atropine at 15 minute intervals until atropinization occurs (tachycardia, flushed skin, dry mouth, mydriasis); Children under 12 - initial dose = 0.05 mg/kg body weight and repeat dose = 0.02 - 0.05 mg/kg body weight. Use of oximes such as 2-PAM is controversial. Observe patient to insure that these symptoms do not recur as atropinization wears off. If in eyes, instill one drop of homatropine. Contains aromatic hydrocarbons that may produce a severe pneumonitis if aspirated during vomiting. Consideration should be given to gastric lavage with an endotracheal tube in place. Treatment is otherwise controlled removal of exposure followed by symptomatic and supportive care.

5 Fire-fighting measures

5.1 Extinguishing media (suitable and not suitable): Foam, CO₂ or dry chemical. Soft stream water fog only if necessary. Contain all runoff.

5.2 Specific hazards:

5.3 Specific methods:

5.4 Protection of firefighters:

6 Accidental release measures

6.1 Personal precautions:

6.2 Environmental precautions:

6.3 Methods for cleaning up:

7 Handling and storage

7.1 Handling

-Technical measures (prevention of user exposure and prevention of fire and explosion):

-Precautions for safe handling of the chemical product:

-Safe handling advice:

7.2 Storage

-Technical measures and storage conditions (e.g. incompatible products and packaging materials):

8 Exposure controls/personal protection

8.1 Engineering measures to reduce exposure:

8.2 Control parameters:

8.3 Personal protective equipment

-Respiratory protection: For splash, mist or spray exposure wear, as a minimum, a properly fitted half-face or full-face air-purifying respirator which is approved for pesticides (U.S. NIOSH/MSHA,

EU CEN or comparable certification organization). Respirator use and selection must be based on airborne concentrations.

-Hand protection: Wear chemical protective gloves made of materials such as nitrile or neoprene. Thoroughly wash the outside of gloves with soap and water prior to removal. Inspect regularly for leaks.

-Eye protection: For splash, mist or spray exposure, wear chemical protective goggles or a face shield.

-Skin and body protection: Depending upon concentrations encountered, wear coveralls or long-sleeved uniform and head covering. For larger exposures as in the case of spills, wear full body cover barrier suit, such as a PVC suit. Leather items - such as shoes, belts and watchbands - that become contaminated should be removed and destroyed. Launder all work clothing before reuse (separately from household laundry).

8.4 Hygiene measures: Clean water should be available for washing in case of eye or skin contamination. Wash skin prior to eating, drinking or using tobacco. Shower at the end of the workday.

9 Physical and chemical properties

9.1 Physical state/from, colour, odour, etc. Aromatic Brown liquid

9.2 PH, with indication of the concentration: 8.0 ± 1.0 (1% solution)

9.3 Specific temperatures/temperature ranges at which change in physical state occur (e.g. boiling point/boiling range): Decomposes on distillation.

9.4 Decomposition temperature: $> 146^{\circ}\text{C}$

9.5 Explosion properties & flash point: $> 56^{\circ}\text{C}$

9.6 Vapour pressure: 0.12 mPa (20°C)

9.7 Vapour density:

9.8 Density: 0.98 (20°C)

9.9 Solubility: With any scale soluble in water and most organic solvents.

9.10 Octanol/water partition coefficient: No available

10 Stability and reactivity

10.1 Conditions to avoid: Excessive heat and fire.

10.2 Material to avoid: Mixing with strongly alkaline or acidic substances should be avoided.

10.3 Hazardous decomposition products:

11 Toxicological information

11.1 Acute toxicity: > 600 mg/kg (rats)

11.2 Local effects:

11.3 Sensitization: Not a skin sensitiser

11.4 Chronic toxicity:

11.5 Long term toxicity: NOEL (2 y) for rats 35 mg/kg diet.

11.6 Specific effects:

11.7 Information according to the different exposure route

-Inhalation: LC_{50} (4 h) for rats > 6.5 mg/l.

-Skin contact: Acute percutaneous LD_{50} for female rats > 3000 mg/kg.

-Eye contact: Slight eye irritation (rabbits).

-Ingestion: ADI (JMPR) 0.03 mg/kg b.w.

12 Ecological information

12.1 Mobility:

12.2 Persistence/Degradability:

12.3 Bioaccumulation: No product of bioaccumulation

12.4 Environmental impact/Ecotoxicity:

13 Disposal considerations

13.1 Waste from residues: Residues in containers should be emptied in a diluted form into a deep pit taking care to avoid contamination of ground waters.

13.2 Contaminated packaging: Container and packaging must either be burned or crushed and buried below topsoil. Care must be taken to avoid subsequent contamination of water sources. Decontamination of containers in order to use them for other purposes should not be permitted.

14 Transport information (Codes and classifications)

14.1 International regulations:

-Land (RID/ADR, DOT 49 CRF):

-International waterways (AND):

-Sea (IMDG-CODE):

-Air (ICAO-TI, IATA-DGRO):

14.2 UN number: UN2991

14.3 Specific precautionary transport measures and conditions:

15 Regulatory information

15.1 Regulations specifically applicable to the chemical product:

15.2 Hazard and safety information as written on the label:

16 Other information

Remark:

1. ADR: European Agreement Concerning the International Carriage of Dangerous:

Goods by Road:

2. RID: Regulations Concerning the International Carriage of Dangerous Goods by Rail:

3. IMDG-CODE: The International Maritime Dangerous Goods Code:

4. IATA: (International Air Transport Association)